



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI CHANDRA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

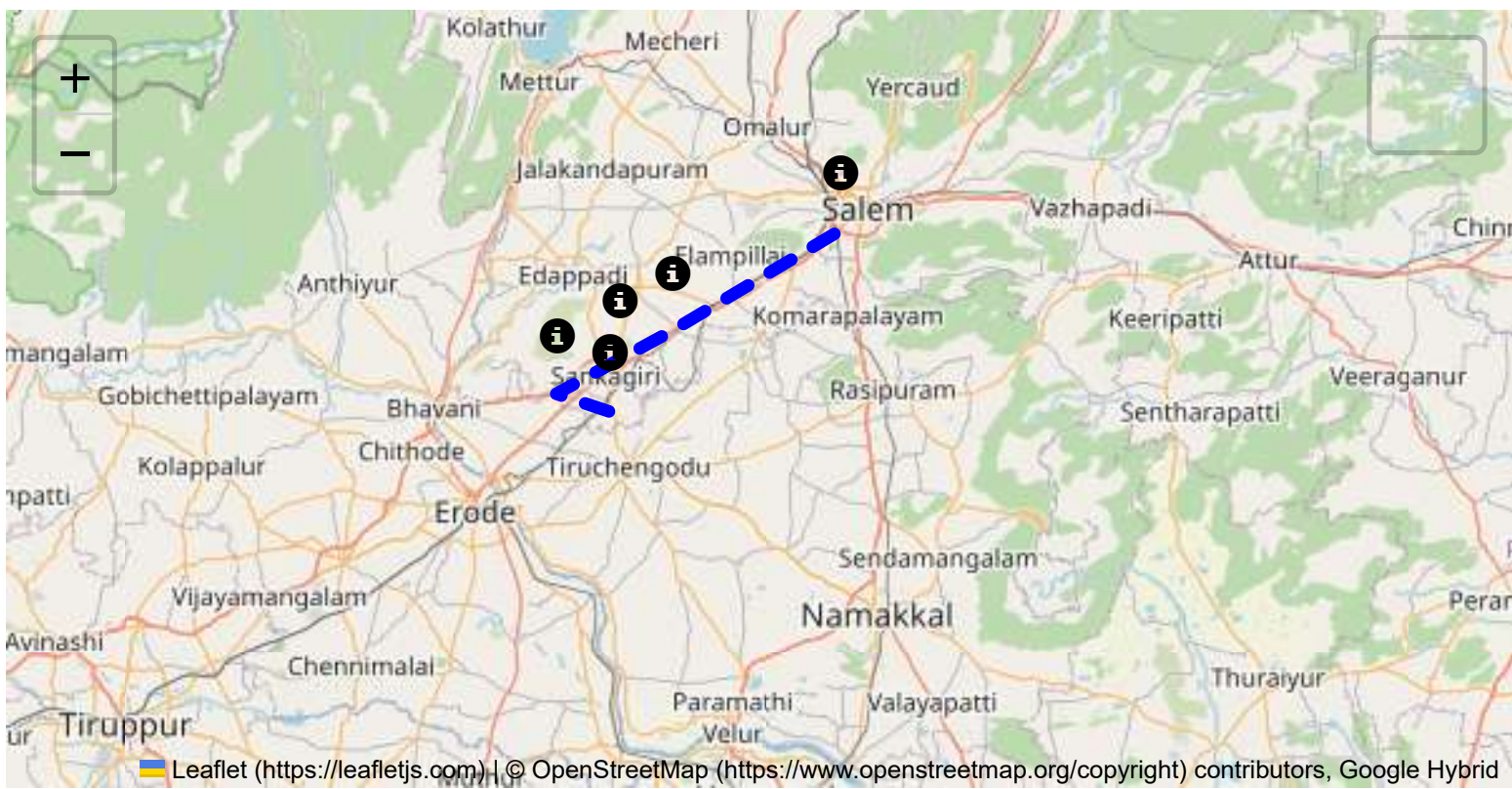
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

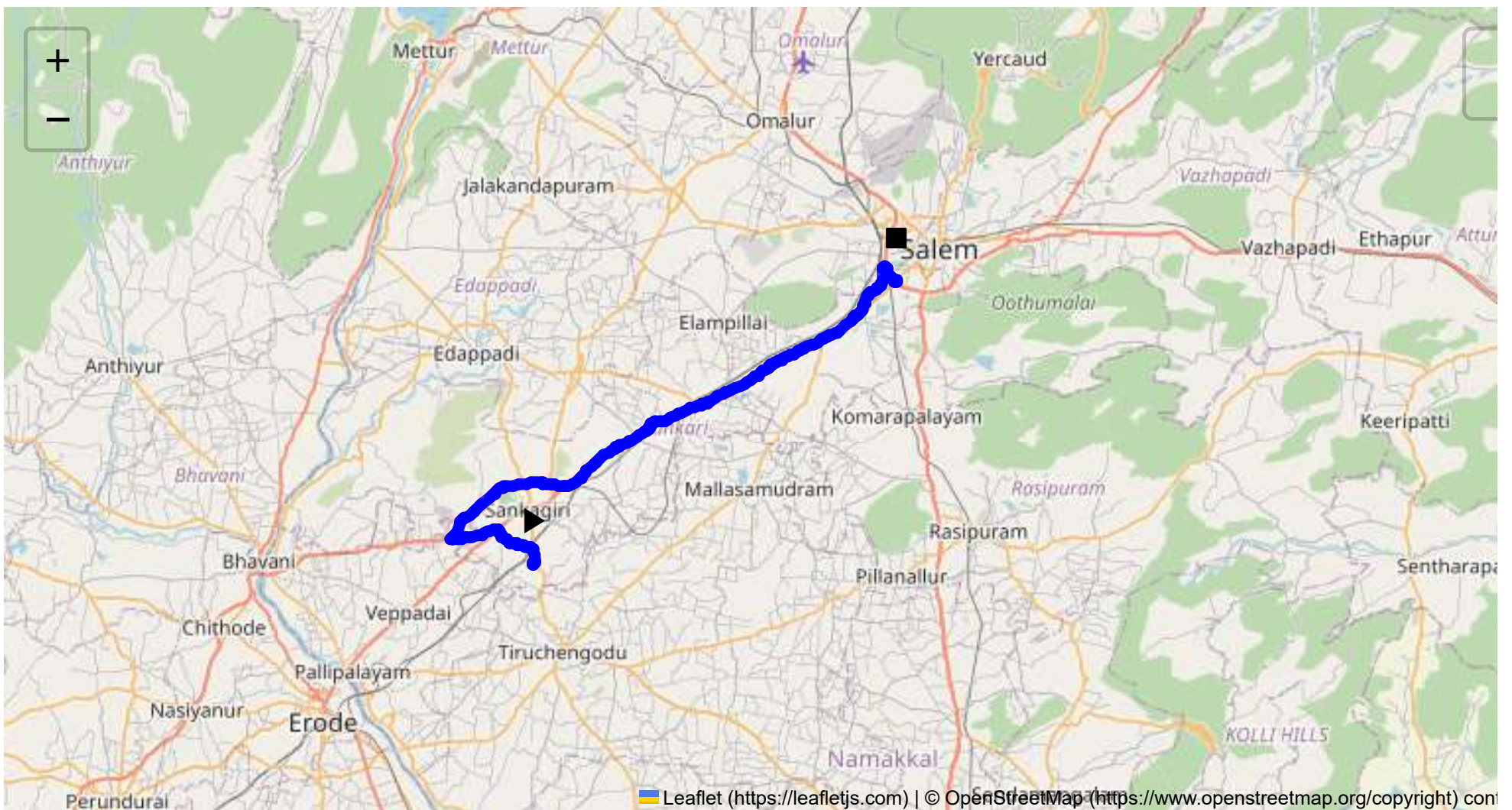
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 51.55 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.631617, 78.126625)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Kondalampatti covers about 51.55 kilometers, connecting rural and urban areas in Tamil Nadu. Key landmarks along the route include Puthur, Padaiveedu, Salem-Kochi Highway, and Vaikundam. The journey typically spans around one hour, with the roads being somewhat narrow and involving highway stretches.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu's climate includes hot summers, moderate monsoons, and mild winters. The monsoon season, from June to September, can result in heavy rainfall, causing waterlogging and making roads slippery. Consider paying special attention to weather forecasts during these months to avoid hazards related to poor visibility and slippery surfaces.

3. Analysis of Traffic Patterns

- **Peak Hours:** Congestion is common during weekday mornings (8:00 AM - 10:00 AM) and evenings (5:00 PM - 7:00 PM), especially near Salem.
- **Congestion-Prone Areas:** The areas around Salem and Sangagiri encounter significant congestion due to both local and through traffic.

4. Assessment of Road Quality and Infrastructure

- **Sangagiri to Puthur:** Generally good road conditions, though rural stretches may have some potholes.
- **Puthur through to Vaikundam:** Includes sections of narrow rural roads; drivers must be cautious of sharp bends and potential livestock on streets.
- **Vaikundam to Salem (Kondalampatti):** The road quality improves as you transition onto the highway approaching Salem, with better signage and lane marking.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** In the event of roadblocks or heavy congestion, consider taking the Tamil Nadu State Highway 20, which parallels the route in some sections.
- Drivers should be familiar with local smaller roads that connect these main highways to manage detours effectively.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Trucking regulations in Tamil Nadu include restrictions on the hours during which hazardous materials can be transported. Transportation is typically restricted during peak urban traffic hours.
- Drivers must be aware of designated routes for hazardous materials to avoid prohibited areas.

7. Overview of Historical Incidents

- There have been occasional incidents involving overturned trucks and chemical spills, specifically in the more industrial areas near Salem. Records of accidents historically indicate a need for cautious speed limits and adherence to safety measures.

8. Environmental Considerations and Sensitive Areas

- The route traverses through some agricultural lands and village areas where environmental protection is important, especially concerning water bodies which might be adversely affected by spills.
- Minimizing noise pollution and air contamination should be prioritized when passing through residential areas.

9. Analysis of Communication Coverage

- Most of the route has reliable communication coverage with occasional weak signals expected in rural stretches between Puthur and Vaikundam.
- Ensuring truck drivers have GPS systems capable of functioning offline can be beneficial.

10. Estimated Emergency Response Times

- **Sangagiri to Puthur:** Estimated response time is around 20-30 minutes due to proximity to local health centers.
- **Vaikundam to Salem:** Can expect quicker response times, closer to 15-20 minutes, due to stronger infrastructure and accessibility to emergency services in urban regions like Salem.

12. Overall Summary of Risk Assessment

This route presents typical risks associated with Indian roads including varied road quality, potential for congestion, and weather-related challenges during monsoon seasons. The adherence to local regulations, awareness of emergency procedures, and strategic planning for alternative routes can mitigate risks substantially. Ensuring proper vehicle maintenance and being mindful of infrastructural variations improve safety for heavy vehicles transporting hazardous materials. Regular updates on traffic conditions and weather, as well as communication reliability checks, are crucial for minimizing risk during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	Medium	11.49377, 77.88573	30 KM/Hr
15	Turn	High	11.49363, 77.88590	15 KM/Hr
16	Turn	High	11.49235, 77.89561	15 KM/Hr
17	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
18	Turn	Medium	11.60942, 78.10056	30 KM/Hr
19	Turn	Medium	11.60965, 78.10070	30 KM/Hr
20	Turn	Medium	11.61736, 78.10588	30 KM/Hr
21	Turn	Medium	11.61745, 78.10601	30 KM/Hr
22	Turn	Medium	11.64158, 78.11808	30 KM/Hr
23	Turn	Medium	11.64098, 78.12163	30 KM/Hr
24	Turn	Medium	11.63266, 78.12756	30 KM/Hr
25	Blind Spot	Blind Spot	11.63185, 78.12654	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
15	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49363, 77.88590



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49235, 77.89561



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60942, 78.10056



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60965, 78.10070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61745, 78.10601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64158, 78.11808



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64098, 78.12163



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63266, 78.12756



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.63185, 78.12654
